Stage 2 Synthesis and Timing Report

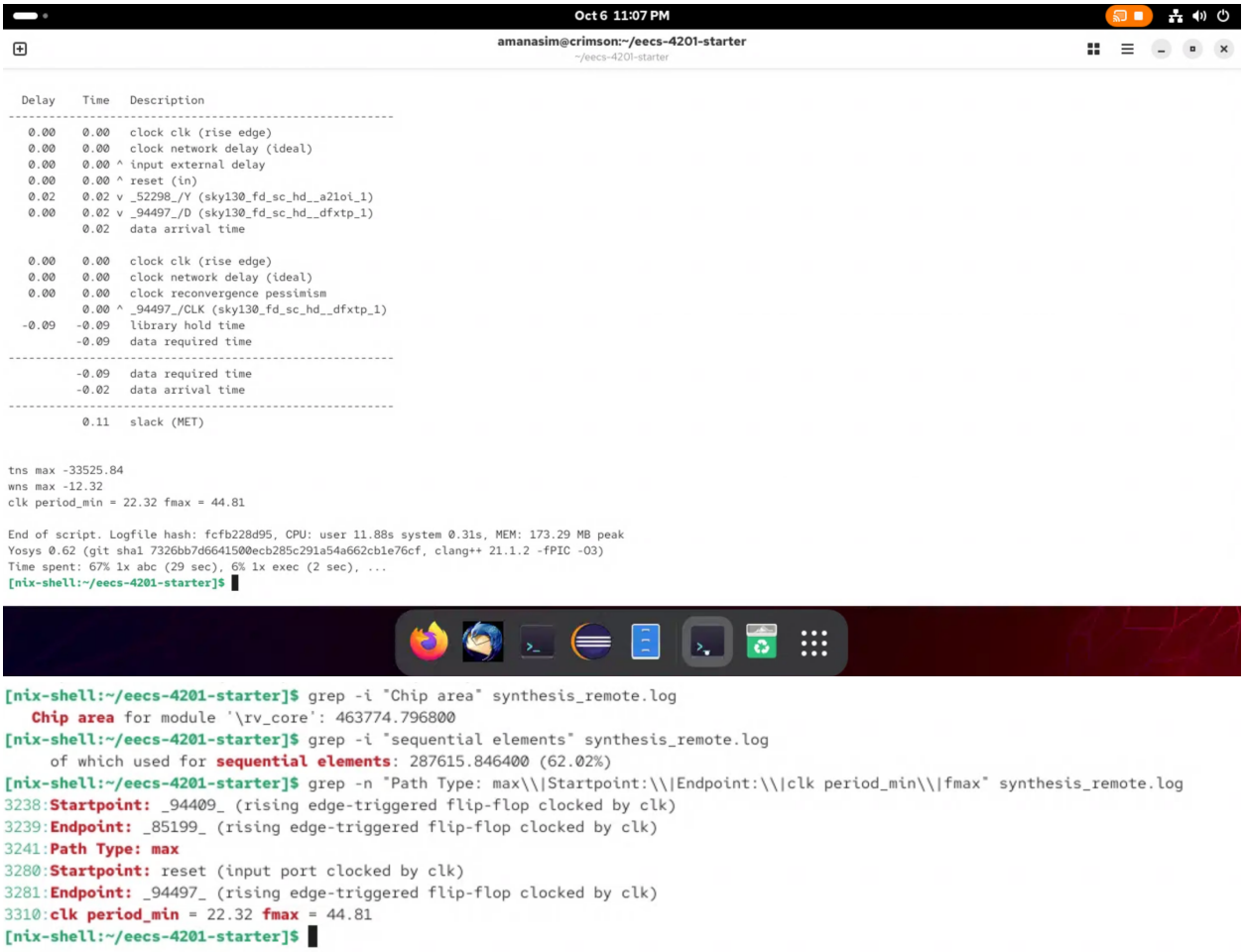
Vanguard-V

The design was synthesized using Yosys and the provided 130 nm standard-cell library. The synthesis completed successfully and generated the required synth.v and synth.json outputs. The final chip area for rv_core was 463774.796800 um^2, with 287615.846400 um^2 used by sequential elements, corresponding to 62.02% of the total area. The large sequential area is expected for a five-stage pipelined processor because pipeline registers, register file state, and other state-holding elements make up a significant part of the design.

For timing / critical path:

Static timing analysis reported a minimum clock period of 22.32 ns, corresponding to an estimated maximum clock frequency of 44.81 MHz. The reported maximum-delay critical path starts at internal flip-flop _94409_ and ends at internal flip-flop _85199_. Since these are synthesized internal cell names, the exact RTL signal path is not directly visible from the summary output, but the path represents the longest register-to-register delay after technology mapping. The timing report also shows WNS max = -12.32 and TNS max = -33525.84, indicating that the design does not meet a 100 MHz clock period, but can be safely clocked at the reported estimated frequency.

Yosys Synthesis Area and Timing Results




```
[nix-shell:~/eecs-4201-starter]$ grep -i "Chip area" synthesis_remote.log
   Chip area for module '\rv_core': 463774.796800
[nix-shell:~/eecs-4201-starter]$ grep -i "sequential elements" synthesis_remote.log
     of which used for sequential elements: 287615.846400 (62.02%)
[nix-shell:~/eecs-4201-starter]$ grep -n "Path Type: max\\|Startpoint:\\|Endpoint:\\|clk period_min\\|fmax" synthesis_remote.log
3238:Startpoint: _94409_ (rising edge-triggered flip-flop clocked by clk)
3239:Endpoint: _85199_ (rising edge-triggered flip-flop clocked by clk)
3241:Path Type: max
3280:Startpoint: reset (input port clocked by clk)
3281:Endpoint: _94497_ (rising edge-triggered flip-flop clocked by clk)
3310:clk period_min = 22.32 fmax = 44.81
[nix-shell:~/eecs-4201-starter]$
```